

MA 2320 Final Exam Sample 3

Name: _____

No calculators, no phones, no electronics allowed. Must show work for credit.

1. Approximate, using $n=4$ rectangles, the value of

$$\int_1^4 x^2 dx$$

2. Compute the area between the graphs of $y = x^2$, and $y = 4 - x^2$.
3. Find the volume of the solid formed by revolving the region bounded by the curves $x = y^2$, the y -axis ($x = 0$), and the horizontal lines $y = 1$ and $y = 2$ about the y -axis.
4. Compute the arclength of $y = x^{3/2}$ from $x = 0$ to $x = 1$?
5. $\int e^x \sqrt{e^x - 1} dx$
6. $\int x \arctan(x) dx$
7. $\int \cos^3(4x) \sin^2(4x) dx$
8. $\int \frac{1}{x^2 \sqrt{1+x^2}} dx$
9. $\int \frac{2x^2 - 2x + 1}{x(x-1)^2} dx$
10. For the following determine if the series converges or diverges. For all of these, as in class, show the following: 1. test/series type, 2. Conclusion 3. Criteria 4. work.

(a) $\sum_{n=1}^{\infty} \frac{4^n}{5^n}$

(b) $\sum_{n=1}^{\infty} \frac{1}{n^3 + n}$

(c) $\sum_{n=1}^{\infty} \frac{n^2}{2^n}$

(d) $\sum_{n=1}^{\infty} \left[\frac{n+1}{n+1} \right]^n$

11. Find the interval of convergence and the radius of convergence for

$$\sum_{n=1}^{\infty} \frac{(x+2)^n}{n}$$

12. Find the Taylor/Maclaurin series from the definition for $f(x) = \cos 4x$.

13. Find the Taylor/Maclaurin series from a known series for $f(x) = \frac{\cos(x)-1}{x^2}$.

14. Graph $x - y^2/4 = 4$.

15. Graph the following equation given in parametric form. $x = 2 \cos(t)$ $y = 3 \sin(t)$.

16. Translate the functions from rectangular into a polar equation.

(a) $x^2 + y^2 y = 1$.

(b) $\tan(\sqrt{x^2 + y^2}) = \frac{y}{x}$.

17. Translate the functions from polar to into a rectangular equation

(a) $r = 4 \sin(\theta)$.

(b) $r^2 = r \cos \theta$.